

Education

- 2024–present **PhD, Mathematics**, *University of Wisconsin-Madison*, WI, United States.
Advisor: Autumn Kent
- 2022–2024 : **VISP Exchange & Master Program of Mathematics**, *University of Wisconsin-Madison*, WI, United States.
- 2019–2023 : **Bachelor of Mathematics and Applied Mathematics**, *Southern University of Science and Technology*, Shenzhen, China.
Advisor: Yannan Qiu

Papers

- 2026 **Counting surface subgroups in cusped hyperbolic 3-manifolds**, Xiaolong Hans Han, Zhenghao Rao, Jia Wan.
arXiv preprint arXiv:2602.20098
- 2023 **Classification of simplicial Coxeter groups in n-dimensional Euclidean, spherical and hyperbolic spaces**.
Undergraduate thesis

Conferences and Talks

- July 24, 2026 **Seminar Talk**, *Counting surface subgroups and atoroidal surface bundles*.
Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS)
- July 13-17, 2026 **Lightning Talk**, *Counting surface subgroups and atoroidal surface bundles*.
ICM satellite conference: Vanderbilt Colloquium Series (VCS) on Groups, Geometry and Dynamics
- June 5-7, 2026 **Short Talk (20min)**, *Counting surface subgroups in a cusped hyperbolic 3-manifold and atoroidal surface bundles*.
Buckeye Metric Geometry Workshop II, University of Toledo
- May 1-3, 2026 **Participation**, Cornell Topology Festival, Ithaca NY.
- April 24-26, 2026 **Poster**, *Counting atoroidal surface bundles*.
Graduate Student Topology and Geometry Conference (GSTGC) 2026, Rutgers University, New Brunswick
- April 20, 2026 **Seminar Talk**, *Counting surface subgroups in cusped hyperbolic 3-manifolds and atoroidal surface bundles*.
University of Wisconsin-Madison
- April 13, 2026 **Seminar Talk**, *Counting surface subgroups in cusped hyperbolic 3-manifolds and atoroidal surface bundles*.
Purdue University
- May 1-3, 2026 **Participation**, Rice Dynamics and Geometry Conference.
- January-March, 2026 **Participation**, SLMATH, Berkeley: Topological and Geometric Structures in Low Dimensions.

- October 4-5, **Lightning Talk**, *Counting surface subgroups in a cusped hyperbolic 3-Manifold*.
2025 Computational problems about 3-manifolds, associated groups and varieties, Rutgers University, Newark
- June 2-6, **Participation**, Midwest Summer School in Geometry, Topology, and Dynamics.
2025 University of Wisconsin-Madison
- May 19-22, **Participation**, The 53rd John H. Barrett Memorial Lectures: Recent Development in Geometric
2025 Group Theory.
The University of Tennessee, Knoxville
- May 5-9, **Participation**, Advanced School on Mapping Class Groups, Surface Subgroups, Bundles, and
2025 Group Extensions.
Instituto de Ciencias Matemáticas (ICMAT), Madrid
- July 10-12, **Participation**, Building Relationships for an Inclusive and Diverse Group of Emerging Students
2024 (BRIDGES).
Salt Lake City, Utah

Fellowships & Awards

- Fall 2021: **The Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)**, China.
National First Prize
- Fall **The Thirteenth/Twelfth Chinese Mathematics Competitions (CMC)**, China.
2021/2020: First Prize in Guangdong Province
- Fall **Excellent Student Scholarship**, Southern University of Science and Technology.
2021/2020: Third Prize

Computer skills

Programming C, C++, JAVA, Python
Languages

Position of Responsibility

- Fall 2026: **Madison Experimental Mathematics Lab (MXM)**, Graduate Mentor.
Building virtually fibered surfaces

Teaching Experience

University of Wisconsin-Madison

- Fall 2026: **Math 320: Linear Algebra and Differential Equations**, Teaching Assistant.
- Fall 2025: **Math 211: Survey of Calculus I**, Teaching Assistant.
- Fall 2024: **Math 221: Calculus and Analytic Geometry I**, Teaching Assistant.
- Spring 2024: **Math 742: Abstract Algebra II**, Grader.
- Fall 2023: **Math 113: Trigonometry**, Grader.